EE 5239 Optimization Homework 2 Cover Sheet

Instructor name: Mingyi Hong Student name: _____

- Date assigned: Wednesday 9/30/2026
- Date due: Friday 10/09/2026 11:59pm
- This cover sheet must be signed and submitted along with the homework answers on additional sheets.
- By submitting this homework with my name affixed above,
  - I understand that late homework will not be accepted,
  - I acknowledge that I am aware of the University's policy concerning academic misconduct (appended below),
  - I attest that the work I am submitting for this homework assignment is solely my own, and
  - I understand that suspiciously similar homework submitted by multiple individuals will be reported to the Dean of Students Office for investigation.
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1 Reading

- Reading: Textbook Section 1.1 - 1.2
- Appendix A.

2 Problems

2.1 Textbook Problems

1. 1.2.1

   Consider the problem of minimizing the function of two variables $f(x, y) = 3x^2 + y^4$

   (a) Apply one iteration of the steepest descent method with $(1, -2)$ as the starting point and with the stepsize chosen by the Armijo rule with

   Steepest Descent:

$$x^{r+1} = x^r - \alpha_r \nabla f(x^r)$$

   $s = 1, \sigma = 0.1, \beta = 0.5$

$$\nabla f(x, y) = \begin{bmatrix} 6x \\ 4y^3 \end{bmatrix}$$

$$\nabla f(1, -2) = \begin{bmatrix} 6 \\ -32 \end{bmatrix}$$

   Steepest direction:

$$d_0 = -\nabla f = \begin{bmatrix} -6 \\ 32 \end{bmatrix}$$

$$f(1, -2) = 19$$

   Armijo rule:

$$f(x_0 + \alpha d_0) \leq f(x_0) + \sigma \alpha \nabla f(x_0)^T d_0$$

$$\nabla f(x_0) d_0 = -\|\nabla f(x_0)\|^2 = -36 - 1024 = -1060$$

$$f(x_0) + \sigma \alpha \nabla f(x_0)^T d_0 = 19 - 106\alpha$$

   Armijo step sizes:

$$\alpha = s\beta^m$$

Therefore we test:

$$\alpha = 1, 1/2, 1/4, 1/8, 1/16...$$

For $m = 4$:

$$\alpha = 1/16 \rightarrow (x, y) = (x_0 + \alpha d_0) = (5/8, 0)$$

$$f(5/8, 0) = 3(5/8)^2 = 1.171875 \; and \; 19 - 106(1/16) = 12.375$$

Since

$$1.171875 \leq 12.375$$

$$\alpha_0 = 1/16$$

The Armijo iterations were done using a python script in my Github

After one iteration:

$$x_1 = (5/8, 0)$$

(b) Repeat (a) using $s = 1, \sigma = 0.1, \beta = 0.1$ instead. How does the cost of the new iterate compare to that obtained in (a)? Comment on the tradeoffs involved in the choice of α

$$\nabla f(x, y) = \begin{bmatrix} 6x \\ 4y^3 \end{bmatrix}$$

$$\nabla f(1, -2) = \begin{bmatrix} 6 \\ -32 \end{bmatrix}$$

Steepest direction:

$$d_0 = -\nabla f = \begin{bmatrix} -6 \\ 32 \end{bmatrix}$$

$$f(1, -2) = 19$$

Armijo rule:

$$f(x_0 + \alpha d_0) \leq f(x_0) + \sigma \alpha \nabla f(x_0)^T d_0$$

$$\nabla f(x_0)^T d_0 = -\|\nabla f(x_0)\|^2 = -1060$$

$$f(x_0) + \sigma \alpha \nabla f(x_0)^T d_0 = 19 - 106\alpha$$

For $m = 1$:

$$\alpha = 0.1 \rightarrow (x, y) = (x_0 + \alpha d_0) = (0.4, 1.2)$$

$$f(0.4, 1.2) = 3(0.4)^2 + (1.2)^4 = 2.5536$$

$$19 - 106(0.1) = 8.4$$

Since

$$2.5536 \leq 8.4$$

$$\alpha_0 = 0.1$$

is accepted.

After one iteration:

$$x_1 = (0.4, 1.2)$$

This cost much less as m had to only be iterated to 1 instead of 4 as in (a). The choice of σ here make it less iterations but the length of the resulting f is greater.

(c) Apply one iteration of Newton's method with the same starting point and stepsize rule as in (a). How does the cost of the new iterate compare to that obtained in (a)? How about the amount of work involved in finding the new iterate?

Newtons method:

$$x_{r+1} = x_r - \alpha_r(\nabla^2 f(x_r))^{-1} \nabla f(x_r)$$

Gradient at starting point (found previously):

$$\nabla f(1, -2) = \begin{bmatrix} 6 \\ -32 \end{bmatrix}$$

Hessian:

$$\nabla^2 f(x, y) = \begin{bmatrix} 6 & 0 \\ 0 & 12y^2 \end{bmatrix}$$

$$\nabla^2 f(1, -2) = \begin{bmatrix} 6 & 0 \\ 0 & 48 \end{bmatrix}$$

Newton direction:

$$d_0 = -(\nabla^2 f(x_0))^{-1} \nabla f(x_0) = - \begin{bmatrix} 1/6 & 0 \\ 0 & 1/48 \end{bmatrix} \begin{bmatrix} 6 \\ -32 \end{bmatrix} = \begin{bmatrix} -1 \\ 2/3 \end{bmatrix}$$

Armijo step size:

$$\alpha = s\beta^m$$

For $m = 0, \alpha = 1$:

$$x_0 + \alpha d_0 = (1, -2) + (-1, 2/3) = (0, -4/3)$$

$$f(0, -4/3) = 256/81$$

$$f(x_0) + \sigma\alpha\nabla f(x_0)^T d_0 = 244/15$$

$$f(x_0 + \alpha d_0) = 256/81 \leq f(x_0) + \sigma\alpha\nabla f(x_0)^T d_0 = 244/15$$

This step is accepted. After one Newton iteration:

$$x_1 = (0, -4/3)$$

Newton's method requires additional calculations for the hessian but converges more rapidly.

2. 1.2.2

Describe the behavior of the steepest descent method with constant stepsize s for the funtion $f(x) = \|x\|^{2+\alpha}$ where $\alpha \geq 0$. For which values of s and x^0 does the method converge to $x^* := 0$. Relate your answer to the assumptions of Prop. 1.2.3.

Prop 1.2.3 (Convergence for a Constant Stepsize): Let x_k be a sequence generated by a gradient method $x^{k+1} = x^k + \alpha^k d^k$ where d^k is a gradeint related. Assume that for some constant $L > 0$, we have

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in R^n$$

and that for all k we have $d^k \neq 0$ and

$$\epsilon \leq a^k \leq (2 - \epsilon)\overline{\alpha}$$

where

$$\overline{\alpha} = \frac{|\nabla f(x^k)' d^k|}{L\|d^k\|^2}$$

and ϵ is a fixed positive scalar. Then every limit point of x^k is a stationary point.

Steepest descent with stepsize $s > 0$:

$$x_{k+1} = x_k - s\nabla f(x_k)$$

$$f(x) = \|x\|^{2+\alpha} = (x^T x)^{\frac{2+\alpha}{2}}$$

Gradient

$$\nabla f(x) = \frac{(2+\alpha)}{2}(x^T x)^{\frac{2+\alpha}{2}-1}\nabla(x^T x) = (2+\alpha)\|x\|^{\alpha}x$$

$$x_{k+1} = x_k - s(2+\alpha)\|x_k\|^{\alpha}x_k = [1 - s(2+\alpha)\|x_k\|^{\alpha}]x_k$$

Norms:

$$\|x^{k+1}\| = |1 - s(2+\alpha)\|x^k\|^{\alpha}|\|x^k\|$$

To converge to 0 the norm must decrease (say to less than 1):

$$|1 - s(2+\alpha)\|x^k\|^{\alpha}| < 1$$

$$-1 < 1 - s(2+\alpha)\|x^k\|^{\alpha} < 1$$

$$s(2+\alpha)\|x^k\|^{\alpha} < 2$$

Case 1: $\alpha = 0$

$$f(x) = \|x\|^2$$

$$x_{k+1} = (1-2s)x^k$$

$$x_k = (1-2s)^k x_0$$

Converges to 0 if:

$$|1-2s| < 1 \; when \; 0 < s < 1$$

So if $\alpha = 0$ then $0 < s < 1$

Case 2: $\alpha > 0$

$$r_k = \|x_k\|$$

$$r_{k+1} = |1 - s(2+\alpha)r_k^{\alpha}|r_k$$

Found earlier:

$$s(2+\alpha)\|x^k\|^\alpha < 2$$

$$s < \frac{2}{(2+\alpha)\|x_0\|^\alpha}$$

$$|1 - s(2+\alpha)r_0^\alpha| < 1$$

$$0 \geq r_{k+1} < r_k$$

Threfore r converges to zero so x converges to zero

For $\alpha = 0$ the gradient is:

$$\nabla f(x) = 2x$$

This is globally Lipschits with $L = 2$

For $\alpha > 0$ the Hessian grows without bound as $\|x\| \to \inf$ making it not globally Lipschitz. Therefore Prop 1.2.3 does not gurantee global convergence with constant step size for all intial points.

3. 1.2.3

Consider the funtion $f : R^n \to R$ given by

$f(x) = \|x\|^{3/2} = (\|x\|^2)^{3/4}$

and the method of steepest descent with a contant stepsize. Show that for this function, the Lipschitz condition $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all x and y is not satisfied for any L. Furthermore, for any value of constant stepsize, the method either converges in finiste number of iterations to the minimizing point $x^* = 0$ or else it does not converge to x^*.

Gradient:

$$f(x) = (\|x\|^2)^{3/4} = u^{3/4}$$

$$\nabla f(x) = \frac{df}{du}\nabla u = 3/4u^{-1/4}\nabla u$$

$$\nabla u = \nabla(x^T x) = 2x$$

$$\nabla f(x) = 3/4(x^T x)^{-1/4}(2x) = 3/2(x^T x)^{-1/4}x = 3/2(\|x\|^{-1/2})x$$

Lipschitz:

Suppose there is an $L > 0$ s.t.:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y$$

Suppose $y = 0$ and x is positive:

$$\|\nabla f(x) - \nabla f(0)\| = \left\| \frac{3}{2}\|x\|^{-1/2}x \right\| = \frac{3}{2}\|x\|^{1/2}$$

Liphshits requires:

$$3/2\|x\|^{1/2} \leq L\|x\|$$

$$3/2 \leq L\|x\|^{1/2}$$

As $x \to 0$

$$L\|x\|^{1/2} \to 0 \leq 3/2$$

So Lipschitz does not apply $\forall L$

Part 2: Steepest descent with constant stepsize

Let the constant stepsize be $\alpha \geq 0$

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

$$x_{k+1} = x_k - \frac{\alpha 3}{2}\|x_k\|^{-1/2}x_k$$

$$x_{k+1} = \left(1 - \frac{3\alpha}{2\sqrt{\|x_k\|}}\right) x_k$$

$$\|x_{k+1}\| = |1 - \frac{3\alpha}{2\sqrt{\|x_k\|}}| \ \|x_k\|$$

Define:

$$c = \frac{3\alpha}{2}, \ r_k = \|x_k\|$$

$$r_{k+1} = |1 - \frac{c}{\sqrt{r_k}}|r_k = |r_k - c\sqrt{r_k}|$$

Define:

$$u_k = \frac{\sqrt{r_k}}{c}$$

$$r_k = c^2 u_k^2$$

Then

$$u_{k+1} = \sqrt{u_k|u_k - 1|}$$

For

$$0 < u_k < 1/2$$

:

$$u_{k+1} = \sqrt{u_k(1 - u_k)}$$

$$u_{k+1}^2 = u_k(1 - u_k) > u_k$$

Therefore as

$$u_k \to 0$$

f begins to move away from 0.

$$r_k = c^2 u_k^2 = \|x_k\|$$

So if u_k does not converge to zero then x_k does not either.

4. 1.2.7

   An electrical engineer wants to maximize the current I between two points A, B of a complex network by adjusting the values x_1, x_2 of the two variable resitors, where $0 \leq x_1 \leq R_1,\ 0 \leq x_2 \leq R_2$ and R_1, R_2 are given. The engineer does not have an adequate mathematical model of the network decides to adpot the following procedure. She keeps the value x_2 of the second resistor fixed and adjust the value of the first resistor until the current I is maximized. She then keeps the value of x_1 of the first resitor fixed and adjust the value of the second resistor until the current I si maximized. She then repeats the procedure unil no further progress can be made. She knows a priori that during this procedure the values $x1, x2$ can never reach their extreme values $0, R1, R2$. Explain whether there is a sound theoretical basis for the engineers procedure. Hint: Consider how the steepest descent method works for two-dimensional problems.

   This does not work. She is using coordinate ascent.

   Since the resistor can never reach their bounds she will reach a point were

   $$\frac{\delta I}{\delta x_1}, \frac{\delta I}{\delta x_2} = 0$$

   This is therefore a stationary point. However a stationary point does not gurantee a global minimum as it could be a saddle point. Then moving in a diagonal direction would further decrease the function.

5. Show that a second-order continuously differentiable function $f(x) : \mathbb{R} \to \mathbb{R}$ is convex *if and only if* its second-order derivative is non-negative.

   $$f \textit{ is convex} \leftrightarrow f"(x) \geq 0,\ \forall x \in \mathbb{R}$$

If $f"(x) \geq 0$ then $f'(x) \leq f'(y)$ for $x < y$

Mean Value theoreom: $\exists\ c \in (x, y) s.t.$:

$$\frac{f(y) - f(x)}{y - x} = f'(c)$$

As f' is not decreasing:

$$f'(x) \leq f'(c)$$

So the tangent line at every

$$f(y) = f(x) + f'(c)(y - x) \geq f(x) + f'(x)(y - x)$$

As it holds for $\forall (x, y)$ then (x, y) can be flipped

$$f(x) \geq f(y) + f'(y)(x - y)$$

Adding $f(x), f(y)$ inequalities together we get:

$$0 \geq f'(x)(y - x) + f'(y)(x - y)$$

$$y - x \geq 0$$

$$f'(x) \leq f'(y)$$

Therefor f' is nondecreasing at all x

$$f"(x) \geq 0$$

6. Suppose that $f(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$ is convex. Please show the following properties

   (a) $f(\mathbf{x}) - f(\mathbf{y}) \geq \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle,\ \forall\ \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$
   Assume f is twice differntiable and convex:

   $$\nabla^2 f(x) \geq 0$$

   Second order taylor expasion:

   $$c = y + t(x - y)$$

   $$f(x) = f(y) + \nabla f(y)^T (x - y) + 1/2 (x - y)^T \nabla^2 f(c)(x - y)$$

   Because f is convex:

   $$\nabla^2 f(c) \geq 0 \to f(x) \geq f(y) + \nabla f(y)^T (x - y)$$

$$f(x) - f(y) \geq \langle \nabla f(y), x - y \rangle$$

x, y were left arbitrary so this applies $\forall x, y \in \mathbb{R}^n$

(b) If $g(y) : \mathbb{R} \to \mathbb{R}$ is a monotonically non-decreasing **convex** function, then $g(f(\mathbf{x}))$ is also a convex function.

$$h(x) = g(f(x))$$

Prove h is convex:
As f is convex:

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

Because g is always increasing:

$$g(f(tx + (1-t)y)) \leq g(tf(x) + (1-t)f(y))$$

Because g is convex (same as done in for f):

$$g(tf(x) + (1-t)f(y)) \leq tg(f(x)) + (1-t)g(f(y))$$

Combining the two inequalities:

$$g(f(tx + (1-t)y)) \leq tg(f(x)) + (1-t)g(f(y))$$

$$h(tx + (1-t)y) \leq th(x) + (1-t)h(y)$$

This defines $h(x) = g(f(x))$ as convex

(c) If $g(y) : \mathbb{R} \to \mathbb{R}$ is a decreasing **convex** function, then provide an example showing that $g(f(\mathbf{x}))$ may not be a convex function.
Since g is decreasing, this does not hold anymore:

$$g(f(tx + (1-t)y)) \leq g(tf(x) + (1-t)f(y))$$

Therefore this is not necessairly true:

$$g(f(tx + (1-t)y)) \leq tg(f(x)) + (1-t)g(f(y))$$

$$h(tx + (1-t)y) \leq th(x) + (1-t)h(y)$$

So h may not be convex

7. Let $\mathbf{A} \neq 0$ be a rank one square matrix, i.e., $\mathbf{A} = \mathbf{x}_0\mathbf{y}_0^T \neq 0$, for some $\mathbf{x}_0, \mathbf{y}_0 \in \mathbb{R}^n$. Consider the following optimization problem

$$\begin{aligned} \min_{\mathbf{x},\mathbf{y}} \quad & \frac{1}{2}\|\mathbf{A} - \mathbf{x}\mathbf{y}^T\|_F^2 \\ \text{s.t.} \quad & \mathbf{x}, \mathbf{y} \in \mathbb{R}^n \end{aligned} \tag{1}$$

Please answer the following questions.

- Show that it is **not** a convex optimization problem.

$$F(x, y) = 1/2\|A - xy^T\|_F^2$$

$$A = x_0 y_0^T \neq 0$$

Two points:

$$(x_1, y_1) = (x_0, y_0),\ (x_2, y_x) = (-x_0, -y_0)$$

First point:

$$x_1 y_1^T = x_0 y_0^T = A$$

$$F(x_1, y_1) = 1/2\|A - A\|_F^2 = 0$$

Second Point:

$$x_2 y_2^T = (-x_0)(-y_0)^T = A$$

$$F(x_2, y_2) = 1/2\|A - A\|_F^2 = 0$$

Midpoint: $(x_{mid}, x_{mid}) = (0, 0)$

$$x_{mid}\ y_{mid}^T = 0$$

$$F(x_{mid}, y_{mid}) = 1/2\|A - 0\|_F^2 = 1/2\|A\|_F^2 > 0\ (As\ A \neq 0)$$

This contradicts convexivity

- Find the set of stationary points.
  Stationary point:

$$\nabla_x F = 0, \nabla_y F = 0$$

Gradients:

$$\nabla_x F = (xy^T - A)y = x(y^T y) - Ay$$

$$A = x_0 y_0^T \rightarrow Ay = x_0(y_0^T y)$$

$$\nabla_x F = x(y^T y) - x_0(y_0^T y)$$

$$y^T y = \|y\|^2 \rightarrow \nabla_x F = x\|y\|^2 - (y_0^T y)x_0$$

Similarly:

$$\nabla_y F = y\|x\|^2 - (x_0^T x)y_0$$

For a stationary point $\nabla_x F = \nabla_y F = 0$
Then:

$$(1)\ \|y\|^2 x = (y_0^T y)x_0,\ (2)\ \|x\|^y 2 = (x_0^T x)y_0$$

Case 1: $x \neq 0,\ y \neq 0$
From 1,2 $x||x_0, y||y_0$. $\alpha, \beta \neq 0$

$$x = \alpha x_0,\ y = \beta y_0$$

Substitute into (1):

$$\|\beta y_0\|^2(\alpha x_0) = (y_0^T \beta y_0)x_0$$

$$\alpha\beta^2\|y_0\|^2 = \beta\|y_0\|^2$$

$$\alpha\beta = 1$$

$$x = \alpha x_0,\ y = \frac{1}{\alpha}y_0,$$

are stationary points
Case 2: $y = 0, x \neq 0$
Equation 1 is automatically satisfied. Equation 2 is:

$$0 = (x_0^T x)y_0 = x_0^T x \to x \perp x_0$$

$$\nabla_x F = \|0\|^2 x - (0)x_0$$

$$\nabla_y F = \|x\|^2(0) - (0)y_0$$

Then $(x, 0), x \perp x_0$ are stationary points
Case 3: $x = 0, y \neq 0$
Equation 2 is automatically satisfied. Equation 1 is:

$$y_0^T y = 0 \to y \perp y_0$$

$$\nabla_x F = \|y\|^2 0 - (0)x_0$$

$$\nabla_y F = \|0\|^2 y - (0)y_0$$

Then $(y, 0), y \perp x_0$ are stationary points

- **(bonus)** Show that every local optimum of the problem is a global optimum

3 Programming Assignment

In this assignment, you are asked to experiment with

1. Steepest descent method with constant step size rule

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

Constant stepsize (Lipschitz):

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$$

$$\alpha = \frac{1}{L} = \frac{1}{\lambda_{max}(Q)}$$

2. Steepest descent method with Armijo step size rule

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

Armijo rule ($s > 0,\ 0 < \beta < 1,\ 0 < \sigma < \frac{1}{2}$)

$$\alpha = s, \beta s, \beta^2 s...$$

$$f(x_k - \alpha \nabla f(x_k)) - f(x_r) \leq \sigma \alpha \nabla f(x_r)^T d_r$$

Faster computing:

$$g_r = Qx_r + b,\ d_r = -g_r$$

Calculate $f(x_r)$ and gd once per step

$$fx = f(x_r),\ gd = g_r^T d_r = -\|g_r\|^2$$

Check each armijo iteration:

$$f(x_r + sd_r) \leq f(x_r) + \sigma s g_r^T d_r$$

3. Diagonally scaled gradient method (using the Hessian diagonals), constant step size rule

$$x_{r+1} = x_r + \alpha d_r$$

$$D = diag(Q_{11}, ..., Q_{nn})$$

$$d_r = -D^{-1} \nabla f(x_r)$$

$$\alpha = \frac{1}{\lambda_{max}(D^{-1/2}QD^{-1/2})}$$

D is the diagonal matrix of the Hessian

4. Conjugate gradient method, exact minimization step size rule

$$x_{r+1} = x_r + \alpha d_r$$

$$g_0 = Qx_0 + b,\ d_0 = -g_0$$

$$\alpha_k = \frac{g_k^T g_k}{d_k^T Q d_k}$$

$$g_{k+1} = Qx_{k+1} + b$$

Fletcher-Reeves:

$$\beta_k = \frac{g_{k+1}^T g_{k+1}}{g_k^T g_k}$$

$$d_{k+1} = -g_{k+1} + \beta_k d_k$$

Repaeat until tolerance is met

to solve the following convex quadratic minimization problem:

$$\begin{aligned} \min \quad & \frac{1}{2}x^T Q x + b^T x \\ \text{s.t.} \quad & x \in \mathbb{R}^n \end{aligned} \tag{2}$$

You can randomly generate the data matrix Q but make sure that it is symmetric and it is positive definite. When condition numbers need to be controlled, note that for any data matrix Q, we have the following eigen decomposition $Q = U\Sigma V$, where U and V are orthonormal matrices and Σ is a diagonal matrix with diagonals being the eigenvalues. You can change the diagonals of Σ to obtain the desired largest and smallest eigenvalues for your data matrix Q.

An example code for perform gradient descent has been provided online at course website, Section 2.2, Example 1 and Example 2.

What you need to turn in:

1. Well documented python codes (or any other codes of a scientific computing language).

2. Plots of CPU time and progression of objective values (relative to the minimum) for various choices of step size rules, algorithms, problem sizes ($n = 100, 500$), and condition numbers ($c = 10, 5000$).

3. Brief discussions of your findings relative to the convergence theory.

   It was found that all four methods converged more slowly for higher condition numbers. This is due to the fact that there are greater differenes in curvature which would make sense that optimization becomes more difficult.

   Steepest descent with a constant stepsize showed a linear convergence which is consistant with theory. Furthermore, greater condition number made the step size smaller because the largest eigenvalue is greater.

   Steepest descent with the armijo rule also had linear convergence with improved convergence when compared to constant stepsize. This however requires addition calculations and therefore CPU times were similar to constant stepsize.

   Diagonlly scaled gradient was not shown to have significantly better CPU time than either steepest descent methods. It was shown to have slightly higher number of iterations than when using the armijo rule and similar iterations when using a constant stepsize.

   Finally, conjugate gradient used substantially fewer iterations than all methods. It theoretically converges in n iterations which was shown to be less than for the other methods. It also performed much better for high condition numbers.

   The size of the problem was shown to increase computational work per iteration but not necessarily iteration count.

   Overall, conjugat gradient was the fastest in terms of iterations and CPU time. Condition number was found to have a significant effect on convergence speed. More sophisticated methods were shown to improve iterations counts but computational costs per iteration remained considerable.